

AN ENDPOINT-SMOOTHED PROLATE VECTOR AND A $\mu^{9/2}(\log \mu)^4$ UPPER BOUND FOR WINDOWED WEIL FORMS

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ABSTRACT. Weil's criterion states that the Riemann Hypothesis holds if and only if Weil's quadratic form W is nonnegative. Restricted to test functions supported in a multiplicative window $[\lambda^{-1}, \lambda]$, this becomes $\lambda_{\min}(\lambda) \geq 0$ for all λ , where $\lambda_{\min}(\lambda)$ is the bottom of the spectrum of the restricted form. Put $\mu = \lambda^2$. In a previous paper we proved, without any hypothesis on the zeros of ζ , that $\lambda_{\min}(\lambda) \leq 5.011 \times 10^{17} \mu^8 (\log \mu)^3 e^{-4\pi\mu}$ for $\mu \geq 50$, using the prolate vector of Connes, Consani and Moscovici. The loss $\mu^{7/2}$ in that bound comes from the jump of the truncated prolate functions at the spatial truncation points $x = \pm\lambda$, which forces the zero sum to be controlled up to height $e^{20\mu}$. We smooth the vector in a layer of width $\asymp c^{-2}$ at these points, where $c = 2\pi\mu$ is the bandwidth. This costs only a bounded factor in the out-of-band mass, and it makes the outer remainder differentiable with a polynomial bound. The zero sum is then dominated by zeros below $3 \cdot 10^{12}$, where RH is verified. We prove, unconditionally,

$$\lambda_{\min}(\lambda) \leq 1.434 \times 10^{23} \mu^{9/2} (\log \mu)^4 e^{-4\pi\mu} \quad (50 \leq \mu \leq 10^4),$$

and $\lambda_{\min}(\lambda) \leq 1.24 \times 10^{23} \mu^5 (\log \mu)^5 e^{-4\pi\mu}$ for all $\mu \geq 50$. The power $9/2$ is that of Fuchs' asymptotic formula for the prolate eigenvalue defect. All constants are explicit and established with ball arithmetic. The results are upper bounds only and say nothing about positivity.

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Status. The proofs passed independent AI reviews with no blocking or major findings; this manuscript was reviewed separately and revised. Constants are established by ball arithmetic (Arb). Not formalized. This paper has not yet been reviewed by independent human experts.

1. INTRODUCTION

1.1. Background. Let W be Weil's quadratic form and W_λ its restriction to test functions supported in the multiplicative window $[\lambda^{-1}, \lambda]$ [1, 2]. Write $\lambda_{\min}(\lambda)$ for the bottom of its spectrum and $\mu = \lambda^2$. By Weil's criterion, RH is equivalent to $\lambda_{\min}(\lambda) \geq 0$ for all λ . Numerically $\lambda_{\min}(\lambda)$ is positive but decays like $e^{-4\pi\mu}$; any approach to RH through these forms must control this decay.

Upper bounds are obtained from test vectors. In [6] we proved $\lambda_{\min}(\lambda) \leq 1.362 \times 10^9 \mu^{23} e^{-4\pi\mu}$ for $\mu \geq 5$. In [5] we analysed the prolate vector k_λ of Connes, Consani and Moscovici [2, §7] and proved (Theorems C and E there)

$$|W(k_\lambda)| \leq P(\mu) \rho_{\text{out}} \|h_\lambda\|^2 + e^{-16\mu} \|h_\lambda\|^2, \quad \lambda_{\min}(\lambda) \leq 5.011 \times 10^{17} \mu^8 (\log \mu)^3 e^{-4\pi\mu} \quad (\mu \geq 50),$$

with $P(\mu) = O(\mu^{7/2} (\log \mu)^3)$. The factor $\rho_{\text{out}} \leq 1 - \lambda_4(2\pi\mu) \leq C \mu^{9/2} e^{-4\pi\mu}$ is controlled by the companion paper [4], so the power of μ in the final bound is $9/2$ plus the power lost in P .

1.2. Where the loss comes from. In [5], h_λ is a combination of prolate functions truncated at $\pm\lambda$. It jumps at the endpoints, so the part r of $\mathcal{E}(h_\lambda)$ outside the window has a Fourier transform decaying only like $1/t$. The zero sum $\sum_\rho |\hat{r}(\gamma_\rho)|^2$ must then be controlled up to height $e^{20\mu}$, far above the height $3 \cdot 10^{12}$ to which RH is verified [8]. Above that height the only available information is the zero-free region [7], of width $\asymp 1/\log T \asymp 1/\mu$, and disc averaging over that width costs μ^3 ; a further $\sqrt{\mu}$ comes from zeros that may lie off the critical line.

1.3. Results. Let ψ_n be the prolate spheroidal wave functions of bandwidth $c = 2\pi\mu$, $h_n(x) = \psi_n(x/\lambda)/\sqrt{\lambda}$ on $|x| < \lambda$, and $I_n = \int h_n$. Let

$$L(u) = (1 - e^{-c^2(1-u^2)/4})^2, \quad D_L = 1 - L, \quad J_n = \int h_n(x) D_L(x/\lambda) dx, \quad \tilde{I}_n = I_n - J_n,$$

and define the *smoothed prolate vector*

$$\tilde{h}(x) = L(x/\lambda)(\tilde{I}_4 h_0(x) - \tilde{I}_0 h_4(x)), \quad \tilde{k} = \mathcal{E}(\tilde{h})|_{[\lambda^{-1}, \lambda]}.$$

Then $\int \tilde{h} = 0$, and $\tilde{h}$ is C^1 with $\tilde{h}(\pm\lambda) = \tilde{h}'(\pm\lambda) = 0$. The factor L differs from 1 only in a layer of width $\asymp c^{-2}$ at the endpoints. As in [5], $h = I_4 h_0 - I_0 h_4$ and $B = \rho_{\text{out}} \|h\|^2 = I_4^2(1 - \lambda_0) + I_0^2(1 - \lambda_4)$ refer to the unsmoothed vector. The quantity B is the natural small parameter of the paper: by [4], $B \leq (1 - \lambda_4(c)) \|h\|^2 \leq 6621 c^{9/2} e^{-2c} \|h\|^2$, and every estimate below is a multiple of B . We continue the theorem lettering of [5], whose main results are Theorems C, D and E.

Theorem F. *Without any hypothesis on the zeros of ζ :*

- (a) for $50 \leq \mu \leq 10^4$, $|W(\tilde{k})| \leq 6.034 \times 10^{14} (\log \mu)^4 B$;
- (b) for every $\mu \geq 50$, $|W(\tilde{k})| \leq P''(\mu) B \leq 5.2 \times 10^{14} \sqrt{\mu} (\log \mu)^5 B$, where P'' is the explicit function (7.1).

Theorem G. *Without any hypothesis on the zeros of ζ :*

- (a) for $50 \leq \mu \leq 10^4$, $\lambda_{\min}(\lambda) \leq 1.434 \times 10^{23} \mu^{9/2} (\log \mu)^4 e^{-4\pi\mu}$;
- (b) for every $\mu \geq 50$, $\lambda_{\min}(\lambda) \leq 1.24 \times 10^{23} \mu^5 (\log \mu)^5 e^{-4\pi\mu}$;
- (c) for every $\mu \geq 50$, $\lambda_{\min}(\lambda) \leq 237522697 P''(\mu) \mu^{9/2} e^{-4\pi\mu}$, and this is less than 0.832 times the bound of [5, Thm. E].

Theorem G(a) improves [5, Thm. E] for $55 \leq \mu \leq 10^4$, by a factor of about 8 at $\mu = 100$ and 4×10^7 at $\mu = 10^4$. The power 9/2 matches that in Fuchs' asymptotic formula $1 - \lambda_4(c) \sim \frac{2^{11}\sqrt{\pi}}{3} c^{9/2} e^{-2c}$ [3] for the prolate eigenvalue defect. We do not claim optimality among Rayleigh-quotient arguments. Theorem G(c) shows that the new bound is never worse than the old one. The cutoff $\mu \leq 10^4$ in part (a) is where we stop absorbing the contribution of the zeros above $3 \cdot 10^{12}$, which grows like a power of μ , into the $(\log \mu)^4$ envelope; it is not optimized.

μ	50	10^3	10^6	10^{12}
Theorem G(a) / [5, Thm. E]	1.27	6.3×10^{-5}	—	—
Theorem G(b) / [5, Thm. E]	30.3	1.2×10^{-2}	4.7×10^{-11}	1.9×10^{-28}
Theorem G(c) / [5, Thm. E]	0.830	2.4×10^{-5}	8.5×10^{-12}	5.8×10^{-30}

TABLE 1. Ratios of the new upper bounds for $\lambda_{\min}$ to the bound $5.011 \times 10^{17} \mu^8 (\log \mu)^3 e^{-4\pi\mu}$ of [5, Thm. E].

1.4. Numerical context.

- With the factor L , the normalized out-of-band mass of $\psi_n L$ ($n = 0, 4$) is 1.20–1.48 times that of ψ_n for $c \in \{10\pi, 14\pi, 20\pi\}$, the factor decreasing in c . For $\mu = 5, 7$ the Galerkin value of W changes by a factor of 1.27–1.32, and $\|\tilde{k}\|^2 / \|\tilde{h}\|^2$ is unchanged to six digits (Section 8).
- The observed ratio $W(\tilde{k})/B$ is about 2 for $\mu = 5, 7$, so the constant in Theorem F is far from sharp.

1.5. Method.

- (1) **The layer.** Its width must be $\asymp c^{-2}$: the prolate functions grow inward from the endpoint like $e^{c\sqrt{2(1-u)}}$, so a layer of width c^{-1} would remove mass $e^{\sqrt{2}c}$ times larger than the endpoint value. With width c^{-2} , the correction is controlled by the endpoint values $|\psi_n(1)|$, which are themselves bounded by $\sqrt{c(1-\lambda_n)}$ (Section 3).
- (2) **Endpoint cancellation.** On the open half-line, $\tilde{r}' = \frac{1}{2}\tilde{r} + \mathcal{E}(x\tilde{h}')$, and $x\tilde{h}'$ is again even with mean zero. Write $x\tilde{h}'$ as the difference of a part coming from the truncated prolate function and a part coming from the layer. Schematically, with $A = |\psi_n(1)|$,

$$\hat{g}^{\text{raw}}(\lambda t) = \frac{2\sqrt{\lambda}\psi'_n(1)}{ct} \sin(ct) + O\left(\frac{\sqrt{\lambda}Ac^2}{t^2}\right), \quad \hat{g}^{\text{lay}}(\lambda t) = \frac{2\sqrt{\lambda}\psi'_n(1)}{ct} \sin(ct) + O\left(\frac{\sqrt{\lambda}Ac^2}{t^2}\right),$$
 so the smoothed function has $\hat{g}^{\text{raw}} - \hat{g}^{\text{lay}} = O(\sqrt{\lambda}Ac^2/t^2)$: the endpoint jump that produced the $1/t$ decay is cancelled exactly. Since $A = O(\sqrt{cB})$, the amplitude is proportional to $\sqrt{B}$ (Section 4).
- (3) **A second boundary regularization.** Smoothing removes the jump inherited from h at $x = \pm\lambda$, but $\tilde{r}$ is still cut at the window boundary $y = -a$, and its value J there can be as large as $\sqrt{cB}$ (Lemma 5.1). A correction $Je^{c^2(y+a)}$, localized at frequency $\asymp c^2$, removes this cut at negligible cost (Section 5).
- (4) **The zero sum.** Zeros below $3 \cdot 10^{12}$ lie on the critical line, and a one-dimensional Sobolev bound weighted by the local zero count gives $(\log \mu)^4$. Higher zeros are handled by the disc averages of [5], now with a t^2 weight that makes their contribution negligible for $\mu \leq 10^4$ (Section 6).

2. SETTING AND INPUTS

2.1. Conventions. We follow [5]: $\lambda > 1$, $\mu = \lambda^2 \geq 50$, $a = \log \lambda$, $c = 2\pi\mu$, $c_0 = 10\pi$, $\ell = 395/399$. Fourier transforms are $\hat{f}(z) = \int f(y)e^{izy}dy$ in the logarithmic variable and $\hat{\varphi}(\xi) = \int \varphi(x)e^{2\pi i x \xi}dx$ in the additive variable. For $n \in \{0, 4\}$ write $\lambda_n = \lambda_n(c)$, $b_n = 1 - \lambda_n$, $A_n = |\psi_n(1)|$, $\chi_{(n)} = \lambda_n^{1/2}$, and $\tilde{\psi}_n$ for the band-limited extension. Let $\chi = \chi_n^{\text{SL}}$ be the Sturm-Liouville eigenvalue of ψ_n , $\sigma = \chi/c^2$, $\mu_n = \chi_{(n)}/\lambda$ (so that $\mu_n \tilde{\psi}_n(t) = \int_{-1}^1 \psi_n(u)e^{icut}du$) and $K_1 = 2 \log(c/3) + 8.08$. For a nontrivial zero ρ of multiplicity m_ρ write $\gamma_\rho = (\rho - \frac{1}{2})/i = t_\rho + i\tau_\rho$, so that t_ρ is the ordinate and $|\tau_\rho| = |\operatorname{Re} \rho - \frac{1}{2}|$; ranges of zeros refer to $|t_\rho|$. Put $H = \|h\|$. For a function f on $(-\infty, -a)$ and $|\tau| < \frac{1}{2}$, $f_\tau(y) = e^{-\tau y}f(y)$; then $\int |\hat{f}(t + i\tau)|^2 dt = 2\pi \|f_\tau\|^2$. Let $\tilde{e} = \mathcal{E}(\tilde{h})$, $\tilde{r} = \tilde{e}\mathbf{1}_{y < -a}$ and $F = \hat{\tilde{r}}$.

$\lambda, \mu = \lambda^2, a = \log \lambda, c = 2\pi\mu$	window, its square, log-width, prolate bandwidth
$\lambda_n = \lambda_n(c), b_n = 1 - \lambda_n, A_n = \psi_n(1) $	prolate eigenvalues, defects, endpoint values
$\lambda_{\min}(\lambda)$	bottom of the spectrum of W_λ
$h, H = \ h\ , B = \rho_{\text{out}}H^2$	unsmoothed vector, its norm, its out-of-band mass
$L, D_L = 1 - L, E_L = e^{-c^2(1-u^2)/4}$	layer factor and its complement
$\tilde{h}, \tilde{k}, \tilde{r}, F = \hat{\tilde{r}}$	smoothed vector, window part, outer part, its transform
$J = \tilde{e}(-a), b_d, q_d = \tilde{r} - b_d$	boundary trace, corrector, corrected remainder

2.2. Results used from earlier papers.

From [5]:

- the radical lemma [5, Lemma 2.2] and the explicit formula [5, (2.1)]: $\hat{\tilde{e}}$ vanishes at every nontrivial zero, so $W(\tilde{k}) = \sum_\rho m_\rho F(\gamma_\rho)F(-\gamma_\rho)$; this holds for any even C^1 function of mean zero supported in $[-\lambda, \lambda]$;
- the Poisson form [5, Lemma 2.3], $\tilde{r}(-\log X) = X^{1/2} \sum_{m \geq 1} \hat{\tilde{h}}(mX) - \frac{1}{2}\tilde{h}(0)X^{-1/2}$ for $X > \lambda$;

- Lemma 3.1 (eigenvalues and pointwise bounds), Lemma 3.2 (Liouville–Green), Lemma 3.3 ($A_n^2 \leq 4cb_n/\lambda_n$), Lemma 3.4 ($h(0)^2 \leq 2.0408\lambda B$), Lemma 3.5 (amplitude), Lemma 3.6 (far field) and Lemma 4.1 ($\|r_\tau\|^2 \leq \lambda^{2\tau}(1-2\tau)^{-1}(592+152.2K_1^2)B$ for $0 \leq \tau < \frac{1}{2}$), all of [5];
- [5, Thm. D]: $\|k\|^2 \geq 0.109H^2$ for $\mu \geq 50$;
- the three external results: the zero count of Trudgian [9], the zero-free region of Mossinghoff and Trudgian [7], and the verification of RH to $3 \cdot 10^{12}$ by Platt and Trudgian [8].

From [4]: $1 - \lambda_4(c) \leq 6621c^{9/2}e^{-2c}$ for $c \geq 10\pi$, hence $B \leq U(c)H^2$ with $U(c) = 6621c^{9/2}e^{-2c}$.

3. THE LAYER

Design of L . The width c^{-2} is the natural scale of the regular singular point $u = 1$ of the prolate equation: by Lemma 3.1, ψ_n stays within a bounded factor of $\psi_n(1)$ on $1 - u \lesssim c^{-2}$, but grows like $e^{c\sqrt{2(1-u)}}$ beyond it. The square in $L = (1 - E_L)^2$ makes both L and L' vanish at $u = \pm 1$. Then $x\tilde{h}'$ vanishes at $\pm\lambda$, and this is exactly what produces the cancellation of the leading terms in Lemma 4.1.

Lemma 3.1 (inward growth). *For $n \in \{0, 4\}$, $c \geq 10\pi$ and $0 \leq x \leq 1$,*

$$\begin{aligned} |\psi_n(1-x)| &\leq A_n e^{2c\sqrt{x}}, & |\psi'_n(1-x)| &\leq c^2 A_n e^{2c\sqrt{x}}, \\ |\psi''_n(1-x)| &\leq 2c^4 A_n e^{2c\sqrt{x}}, & |\psi'''_n(1-x)| &\leq 3c^6 A_n e^{2c\sqrt{x}}. \end{aligned}$$

Proof. Integrating the prolate equation from the regular singular endpoint gives $(1-u)^2\psi'(u) = -\int_u^1 (c^2v^2 - \chi)\psi(v)dv$, and $|c^2v^2 - \chi| \leq c^2$ since $\chi/c^2 < 0.3$ [5, Lemma 3.2]. With $f(x) = \psi(1-x)$ and $x(2-x) \geq x$,

$$|f(x)| \leq A + c^2 \int_0^x \frac{1}{s} \int_0^s |f(t)| dt ds.$$

The positive operator on the right maps t^k to $x^{k+1}/(k+1)^2$, and the remainder of the N -th iterate is at most $\max |f|(c^2x)^N/(N!)^2 \rightarrow 0$. Hence $|f(x)| \leq A \sum_k (c^2x)^k/(k!)^2 \leq Ae^{2c\sqrt{x}}$, using $\binom{2k}{k} \leq 4^k$. The bound for ψ' follows from the same integral identity. For ψ'' and ψ''' , differentiate the regular form $\psi'(u) = -(1+u)^{-1} \int_0^1 [c^2(u+(1-u)v)^2 - \chi]\psi(u+(1-u)v)dv$ and use $c \geq 10\pi$. $\square$

Let $E_L = e^{-c^2(1-u^2)/4}$, so that $D_L = 2E_L - E_L^2$. Then $|D_L| \leq 2E_L$, $|D'_L| \leq c^2 E_L$, $|D''_L| \leq 2c^4 E_L$ and $|D'''_L| \leq 2c^6 E_L$. Substituting $t_1 = c^2(1-u)$ and completing the square,

$$(3.1) \quad \int_0^1 e^{2c\sqrt{1-u}} E_L du \leq c^{-2} \int_0^\infty e^{2\sqrt{t_1}-t_1/4} dt_1 \leq 8e^8 c^{-2}.$$

Similarly, with $d_n = h_n D_L(\cdot/\lambda)$,

$$(3.2) \quad \|d_n\| \leq \frac{\sqrt{32}e^8}{c} A_n, \quad |J_n| \leq \|d_n\|_1 \leq \frac{32e^8\sqrt{\lambda}}{c^2} A_n, \quad \text{TV}(d_n) \leq \frac{2+48e^8}{\sqrt{\lambda}} A_n.$$

Lemma 3.2 (coefficients). $I_0^2 \geq \pi\ell\lambda/c$. With $S = |\tilde{I}_4|A_0 + |\tilde{I}_0|A_4$ we have $S \leq 3\sqrt{cB}$ and $(J_0^2 + J_4^2)^{1/2} \leq z\sqrt{B}/c$, where $z = 32e^8\sqrt{8/\pi}/\ell$.

Proof. ψ_0 is positive in $(-1, 1)$, so $1 = \int \psi_0^2 \leq \|\psi_0\|_\infty \int \psi_0$, and $\|\psi_0\|_\infty \leq \sqrt{c/(\pi\ell)}$ by [5, Lemma 3.1]. By [5, Lemma 3.3] and $b_0 \leq b_4$, $(|I_4|A_0 + |I_0|A_4)^2 \leq 8cB/\ell$ and $A_0^2 + A_4^2 \leq 8cb_4/\ell \leq 8cB/(\ell I_0^2)$. Combining with (3.2) gives the bound for $(J_0^2 + J_4^2)^{1/2}$. Then $S \leq \sqrt{8cB/\ell} + (z\sqrt{B}/c)\sqrt{A_0^2 + A_4^2}$, and $A_0^2 + A_4^2 \leq 8cb_4/\ell \leq 8cU(c)/\ell$, so $S \leq \sqrt{8cB/\ell}[1 + z\sqrt{U(c)}/c]$. The bracket decreases in c ; ball arithmetic at c_0 gives $S \leq 2.84278\sqrt{cB}$. $\square$

Proposition 3.3 (out-of-band mass and norms). Let $\delta h = \tilde{h} - h$ and $\eta(c) = \sqrt{32}e^8\sqrt{8/(\ell c)} + z/c$. Then

- (a) $\|\delta h\| \leq \eta(c)\sqrt{B}$, so the out-of-band mass of $\tilde{h}$ is at most $[1 + \eta(c)]^2 B$;
- (b) $|\tilde{h}(0)|^2 \leq \theta(c)^2 \lambda B$ with $\theta(c) = \sqrt{2.0408} + 2z/(\sqrt{\ell}c)$;
- (c) $\|\tilde{k}\|^2 \geq 0.1089 H^2$ for $\mu \geq 50$;
- (d) $\text{TV}(\delta h) \leq V(c)\sqrt{\lambda B}$, with $V(c)$ given in Appendix A.

Proof. (a) $\delta h = -D_L h + L(-J_4 h_0 + J_0 h_4)$; bound the first term by (3.2) and Lemma 3.2, and the second by orthogonality of h_0, h_4 and $\|L\|_\infty \leq 1$. The orthogonal projection onto out-of-band frequencies is a contraction. (b) [5, Lemma 3.4] and the pointwise bound for $h_n(0)$. (c) Each $n \leq \mu$ contributes $\|e^{y/2} v(n e^y)\|_{[-a, a]}^2 \leq \|v\|^2/(2n)$, so $\|\tilde{k} - k\| \leq \sqrt{2\mu} \|\delta h\| \leq \sqrt{2\mu} \eta \sqrt{U} H$. With [5, Thm. D], $\|\tilde{k}\| \geq (\sqrt{0.109} - \sqrt{2\mu} \eta \sqrt{U}) H$; the correction is below 10^{-124} at $\mu = 50$ and decreasing. (d) $\text{TV}(Lh_n)$ is bounded via [5, Lemma 3.1], $\text{TV}(d_n)$ by (3.2), and the coefficients by Lemma 3.2. $\square$

Since δh is even, of bounded variation and of mean zero, the Riemann-sum error gives $|\mathcal{E}(\delta h)(y)| \leq \text{TV}(\delta h) e^{y/2}$. Combined with [5, Lemma 4.1] (and $\|r_\tau\| \leq \lambda^\tau \|r_0\|$ for $\tau < 0$), for all $|\tau| < \frac{1}{2}$,

$$\|\tilde{r}_\tau\|^2 \leq \frac{\lambda^{2\tau}}{1 - 2\tau} m_0(c) B, \quad m_0(c) = \left[\sqrt{2\{592 + 152.2(2 \log(c/3) + 8.08)^2\}} + V(c) \right]^2 = O((\log c)^2). \quad (3.3)$$

4. HIGH-FREQUENCY FOURIER BOUNDS

Let $g_{n,L}(x) = x(h_n L(\cdot/\lambda))'(x)$, so that $g_1 := x\tilde{h}' = \tilde{I}_4 g_{0,L} - \tilde{I}_0 g_{4,L}$. Write $g_{n,L} = g_n^{\text{raw}} - g_n^{\text{lay}}$ with $g_n^{\text{raw}} = \lambda^{-1/2} u \psi'_n(u)$ and $g_n^{\text{lay}} = \lambda^{-1/2} u(\psi_n D_L)'(u)$, $u = x/\lambda$, on $|u| < 1$.

Lemma 4.1 (cancellation). *For $n \in \{0, 4\}$, $A = A_n$ and $t \geq c$,*

$$|\hat{g}_{n,L}(\lambda t)| \leq C_f \sqrt{\lambda} A \frac{c^2}{t^2}, \quad C_f = 20 + 321e^8.$$

For $2 \leq t < c$, $|\hat{g}_{n,L}(\lambda t)| \leq A_f \sqrt{\lambda} A$ with $A_f = 10 + 48e^8$, and $\int_1^2 |(t\tilde{\psi}_n)'|^2 dt \leq 400c^2 A^2$.

Proof. *The raw part.* Integration by parts gives exactly

$$\hat{g}_n^{\text{raw}}(\lambda t) = -\frac{\chi_{(n)}}{\sqrt{\lambda}} (t\tilde{\psi}_n)'(t) + 2\sqrt{\lambda} \psi_n(1) \cos(ct).$$

Put $w(t) = \sqrt{t^2 - 1} \tilde{\psi}_n(t)$. Then $w'' + (c^2 + \mathcal{V})w = 0$ with $\mathcal{V} = c^2(1 - \sigma)/(t^2 - 1) + (t^2 - 1)^{-2}$, $\sigma = \chi/c^2$. From the finite Fourier representation, $w = C \sin(ct) + O_c(t^{-1})$ with $C = 2\psi_n(1)/(\mu_n c)$ and $|C| \leq A/\sqrt{c}$. Let $w_0 = C[\sin(ct) + \alpha t^{-1} \cos(ct)]$ with $\alpha = -c(1 - \sigma)/2$. The residual $\mathcal{R} = w''_0 + (c^2 + \mathcal{V})w_0$ satisfies $|\mathcal{R}| \leq 2|C|c^3/t^3$ for $t \geq c$. Since $w - w_0$ and its derivative vanish at infinity, the Green formula with kernel $\sin(c(s-t))/c$ gives $|w - w_0|(t) \leq c^{-1} \int_t^\infty |\mathcal{R}| + c^{-1} \int_t^\infty \mathcal{V}|w - w_0|$; with $\int_t^\infty |\mathcal{R}| \leq |C|c^3/t^2$ and $c^{-1} \int_t^\infty \mathcal{V} \leq 2$ for $t \geq c$, Gronwall's inequality gives $|w - w_0| \leq e^2 |C| c^2/t^2$ and $|(w - w_0)'| \leq 6|C|c^3/t^2$. Hence, for $t \geq c$,

$$\left| \hat{g}_n^{\text{raw}}(\lambda t) - \frac{2\sqrt{\lambda} \psi'_n(1)}{ct} \sin(ct) \right| \leq 18\sqrt{\lambda} A \frac{c^2}{t^2},$$

since $\chi_{(n)} C c / \sqrt{\lambda} = 2\sqrt{\lambda} \psi_n(1)$ and $c \alpha \psi_n(1) = \psi'_n(1)$; the $\cos(ct)$ terms cancel.

The layer part. Let $\phi(u) = u(\psi_n D_L)'(u)$. At $u = 1$, $D_L = 1$, $D'_L = 0$ and $D''_L = -c^4/2$, and $\psi'_n(1) = \frac{1}{2}(\chi - c^2)\psi_n(1)$. Lemma 3.1 and (3.1) give $\int_{-1}^1 |\phi''| \leq 321e^8 c^4 A$. Two integrations by parts give

$$\left| \hat{g}_n^{\text{lay}}(\lambda t) - \frac{2\sqrt{\lambda} \psi'_n(1)}{ct} \sin(ct) \right| \leq (2 + 321e^8) \sqrt{\lambda} A \frac{c^2}{t^2}.$$

The two leading terms are identical, and subtracting gives the first claim.

Lower frequencies. For $2 \leq t < c$, use the Liouville–Green representation of [5, Lemmas 3.2 and 3.5]: $|(t\tilde{\psi})'| \leq 3\sqrt{c}A$, and $\|g_n^{\text{lay}}\|_\infty \leq \|g_n^{\text{lay}}\|_1 \leq 48e^8\sqrt{\lambda}A$. For $1 \leq t < 2$, split at $t = 1 + 1/(2c^2)$ and integrate the Liouville–Green bounds; the constant is $320.4 < 400$. $\square$

Proposition 4.2. *For all $|\tau| < \frac{1}{2}$,*

$$\|\mathcal{E}(g_1)\mathbf{1}_{y < -a}\|_\tau^2 \leq \frac{\lambda^{2\tau}}{1 - 2\tau} G(c)B, \quad G(c) = [g_A c^2 + 120c^{3/2}]^2, \quad g_A = \frac{3(A_f + 2C_f)}{\sqrt{2\pi}}.$$

Proof. g_1 is even, continuous, compactly supported, $g_1(0) = 0$ and $\int g_1 = -\int \tilde{h} = 0$. By Lemma 4.1 its Fourier samples are absolutely summable, so $\mathcal{E}(g_1)(-\log X) = \sqrt{X} \sum_{m \geq 1} \hat{g}_1(mX)$ for $X \geq \lambda$. Put $u = X/\lambda$. Split the sum into $mu < c$ (at most c/u terms, each at most $A_f\sqrt{\lambda}S$) and $m \geq \lceil c/u \rceil$, where $\sum_{m \geq M} m^{-2} \leq 2/M \leq 2u/c$. Apart from the band-edge term $m = 1$, $1 \leq u < 2$, this gives $|\mathcal{E}(g_1)(-\log(\lambda u))| \leq \lambda S c (A_f + 2C_f) u^{-1/2}$. The band-edge derivative term is $\sqrt{u} \mathcal{H}(u)$ with $\mathcal{H}(u) = \sum_{n=0,4} \alpha_n \chi_{(n)}(u \tilde{\psi}_n(u))'$, $\alpha_0 = \tilde{I}_4$, $\alpha_4 = -\tilde{I}_0$; by Lemma 4.1 and Minkowski, $\|\mathcal{H}\|_{L^2(1,2)} \leq 20cS$, and $(1 - 2\tau)u^{2\tau} \leq 4$ on $[1, 2]$ gives the term $120c^{3/2}$. Integrate against $u^{2\tau-1}du$ and use $S \leq 3\sqrt{cB}$ and $\lambda^2 = c/(2\pi)$. $\square$

No term of size $H/\sqrt{B}$ appears: every amplitude is a multiple of the endpoint values A_n , which are $O(\sqrt{cB})$.

5. THE BOUNDARY TRACE AND CONDITION (A)

Endpoint smoothing removes the discontinuity inherited from h at $x = \pm\lambda$. The truncation at the window boundary $y = -a$ still leaves a trace J . We remove this remaining singularity with a localized exponential corrector, the second of the two boundary regularizations.

Lemma 5.1 (the trace). *Let $J = \tilde{e}(-a)$. Then $J^2 \leq D_J(c)B \leq (16c + 337392159829(\log c)^2)B$, where $D_J(c) = [\sqrt{8c} + E_J(c)]^2$ and $E_J(c) = O(\log c)$ is given in Appendix A.*

Proof. Apply the Poisson form at $X = \lambda$. The first term is exactly $M(c) = \sqrt{\lambda} \hat{h}(\lambda) = I_4 \chi_{(0)} \psi_0(1) - I_0 \chi_{(4)} \psi_4(1)$, a band-edge value with $|M| \leq \sqrt{8cB}$. For $m \geq 2$, the far-field phase is pure sine: two integrations by parts of the finite Fourier relation give $\tilde{\psi}_n(t) = 2\psi_n(1) \sin(ct)/(\mu_n ct) + O_c(t^{-2})$, so the phase Φ_n of [5, Lemma 3.6] satisfies $\sin \Phi_n = 0$. Hence $|\sum_{m \geq 2} \sin(cm)/m| \leq \pi/2 + 1$, also when $c \in 2\pi\mathbb{Z}$. The phase errors are summed as in [5, Lemma 4.1], the $h(0)$ term is bounded by [5, Lemma 3.4], and the smoothing changes the value by at most $\text{TV}(\delta h)/\sqrt{\lambda} \leq V(c)\sqrt{B}$. $\square$

Remark 5.2. Numerically the trace is not small compared with $\sqrt{B}$: for the smoothed vector, $J^2/B(\tilde{h}) = 7.17, 13.9$ and 20.2 at $\mu = 5, 7, 10$, where $B(\tilde{h})$ is the out-of-band mass of $\tilde{h}$ (Section 8); we do not use or prove an asymptotic statement about J . Lemma 5.1 still does not cost a power of μ in Theorem F, because the corrector b_d below is localized. At the zeros on the verified part of the critical line, its Fourier transform $Je^{-iaz}/(d + iz)$ contributes at most $2J^2S(d) = O(J^2 \log c/c^2)$ to the zero sum. Above that height the factor $\lambda^{2\tau}$ is kept, as in Section 6. For the *unsmoothed* vector we observed that at the first eight zeros $|\hat{k}(t_\rho)|^2$ is only 1.5×10^{-5} – 2.0×10^{-3} times $J^2/(\frac{1}{4} + t_\rho^2)$ (Section 8). A heuristic explanation is that near the band edge the remainder oscillates like a chirp of frequency $\gg c$, so the cut acts like a $1/t$ tail only at high frequency. We have not proved this, and it is not used.

Let $d = c^2$, $b_d(y) = Je^{d(y+a)}\mathbf{1}_{y < -a}$ and $q_d = \tilde{r} - b_d$. Then $q_d(-a) = 0$, $\|(b_d)_\tau\|^2 = \lambda^{2\tau} J^2/(2(d-\tau))$, and $\hat{b}_d(z) = Je^{-iaz}/(d + iz)$. On the open half-line,

$$(5.1) \quad q'_d = \frac{1}{2}q_d + \mathcal{E}(g_1) + (\tfrac{1}{2} - d)b_d.$$

Proposition 5.3 (condition (A)). *For $\mu \geq 50$ and $|\tau| < \frac{1}{2}$, $e^{-\tau y} q_d$ extended by zero lies in $H^1(\mathbb{R})$, and*

$$\|(e^{-\tau y} q_d)'\|^2 \leq \frac{\lambda^{2\tau}}{1-2\tau} K_*(\mu) B, \quad K_* = [\sqrt{Z} + g_A c^2 + 120c^{3/2} + c\sqrt{D_+(c)}]^2 = O(c^4),$$

where $Z = Z_C(\log \mu)^2$, $Z_C = 125363711220$, and $D_+(c) = 16c + 337392159829(\log c)^2$. We write $K_*(\mu)$ throughout.

Proof. By (3.3), Lemma 5.1 and $(1-2\tau)/(2(d-\tau)) \leq 1/d$, $\|(q_d)_\tau\|^2 \leq \lambda^{2\tau}(1-2\tau)^{-1} Z_{\text{raw}} B$ with $Z_{\text{raw}} = [\sqrt{m_0} + \sqrt{D_J}/c]^2 \leq Z$; the last inequality reduces to $\mu = 50$ by monotonicity (Appendix A). Insert this, Proposition 4.2 and Lemma 5.1 into (5.1), using $|\frac{1}{2} - \tau| \leq 1$ and $(d - \frac{1}{2})^2(1-2\tau)/(2(d-\tau)) \leq c^2$. Since $\tilde{h}$ is C^1 and vanishes to first order at $\pm\lambda$, $\tilde{e}$ is C^1 on the open half-line. As $q_d(-a) = 0$, the zero extension has the same weak derivative. $\square$

6. THE ZERO SUM

Write $F = Q + \hat{b}_d$ with $Q = \hat{q}_d$, and $|F|^2 \leq 2|Q|^2 + 2|\hat{b}_d|^2$. Let $T_0 = 3 \cdot 10^{12}$. The zeros are split into three ranges:

range of $ t_\rho $	location of zeros	method	contribution / B
$\leq T_0$	on the critical line [8]	1D Sobolev, local count	$O((\log \mu)^4)$
$(T_0, \max(T_0, c^3)]$	zero-free region [7]	discs, zeroth-order bound	$O(\sqrt{\mu}(\log \mu)^5)$; empty if $\mu \leq 2295$
$> \max(T_0, c^3)$	zero-free region [7]	dyadic discs, t^2 weight	$O(\sqrt{\mu}(\log \mu)^3)$

6.1. Zeros on the verified part of the critical line. For $0 < t_\rho \leq T_0$ we have $\tau_\rho = 0$ [8]. For x in $I_\rho = [t_\rho - \frac{1}{2}, t_\rho + \frac{1}{2}]$, $|F(t_\rho)|^2 \leq |F(x)|^2 + \int_{I_\rho} 2|F||F'|$; averaging over x gives

$$(6.1) \quad \sum_{0 < t_\rho \leq T_0} m_\rho |F(t_\rho)|^2 \leq \frac{1}{2\pi} \int_{\mathbb{R}} \omega(t) (|F|^2 + 2|F||F'|) dt, \quad 0 \leq \omega(t) \leq 12\pi \log(|t| + e + 1),$$

where $\omega/2\pi$ counts zeros within $\frac{1}{2}$ of t , bounded with [9]. With $R = c^3$ we use $\log(|t| + e + 1) \leq \log(2R) + t^2/R^2$. Since $\hat{q}'_d$ is the transform of iyq_d , Plancherel, Proposition 5.3, (3.3) and the bounds $\|yq_d\|^2 \leq 2(a+4)^2 ZB$, $\|(yq_d)'\|^2 \leq YB$ (from $\tau = \frac{1}{4}$) bound the Q part by $48\pi[\mathfrak{A}_0 + 2\sqrt{\mathfrak{A}_0\mathfrak{A}_1}]B$, where $\mathfrak{A}_0 = O((\log \mu)^3)$ and $\mathfrak{A}_1 = O((\log \mu)^5)$ are defined in Appendix A; thus $\sqrt{\mathfrak{A}_0\mathfrak{A}_1} = O((\log \mu)^4)$. The factor 48π is 12π from (6.1), 2 for the two signs of t_ρ , and 2 from $|F|^2 \leq 2|Q|^2 + 2|\hat{b}_d|^2$. For the b_d part, $\sum_{t_\rho > 0} m_\rho/(d^2 + t_\rho^2) \leq S(d) = O(\log d/d)$.

6.2. Zeros above the verified height. We use the disc averages of [5, Lemma 5.1] on dyadic ranges. Let $T \geq T_0$, $x_T = \log(3T)$, $x_j = \log(3 \cdot 2^j T)$ and $r_j = 1/(11.15x_j)$. For $2^j T < |t_\rho| \leq 2^{j+1} T$, take discs of radius $1/(11.15 \log |t_\rho|) \geq r_j$ centred at γ_ρ . By the zero-free region [7] they lie in $|\tau| \leq \frac{1}{2} - r_j$, and Trudgian's count bounds the overlap by $n(x_j)$; zeros of opposite sign never cover the same point, so no extra factor 2 is needed. The weight t^2 and Proposition 5.3 reduce the sum over the j -th range to $\int_{-1/2+r_j}^{1/2-r_j} \lambda^{2\tau}(1-2\tau)^{-1} d\tau$ times explicit factors. We use two bounds for this strip integral. For Theorem F(a) we take $T = T_0$ and bound the integral over the j -th range by $L_j = \frac{1}{2} + \frac{\lambda}{2} \log \frac{11.15x_j}{2}$, as in [5, proof of Lemma 5.1]; this needs no condition on r_j . Since $L_j \leq (x_j/x_0)L_T(\lambda)$ with L_T of Appendix A, this growth is one of the four factors $x_j/x_T \leq 1 + j/40$ absorbed by S_4 below. For Theorem F(b) we take $T = \max(T_0, c^3)$ and use Lemma 6.1. Its hypothesis $r_j \log \mu < 1$ holds: if $c^3 \leq T_0$ then $\mu \leq \mu_*$ (see below) and $\log \mu/(11.15x_T) \leq \log \mu_*/(11.15 \log(3T_0)) < 0.03$,

and if $c^3 > T_0$ then $x_T \geq 3 \log c > 3 \log \mu$, so $\log \mu / (11.15x_T) < 1/33$. Since $r_j \leq r_0 = 1/(11.15x_T)$, the hypothesis holds for every j . Thus, for Theorem F(b) (both cases),

$$\sum_{|t_\rho| > T} m_\rho |Q(\gamma_\rho)|^2 \leq \mathcal{A}(T) \lambda j(x_T, \log \mu) K_*(\mu) B, \quad \mathcal{A}(T) = \frac{2(1000/999)^2 S_4 n(x_T) (11.15x_T)^2}{T^2},$$

with $S_4 = \sum_{j \geq 0} 4^{-j} (1 + j/40)^4 = 572587/414720$ (which absorbs $x_j/x_T \leq 1 + j/40$), and j the strip factor of Lemma 6.1. For b_d the contribution is at most $4\lambda J^2 (\log T + 1)/(\pi T)$, by $|\hat{b}_d(t + i\tau)|^2 \leq \lambda J^2/t^2$ and $\sum_{|t_\rho| > T} m_\rho/t_\rho^2 \leq 2(\log T + 1)/(\pi T)$.

Lemma 6.1 (strip integral). *For $0 < r < \frac{1}{2}$ with $r \log \mu < 1$,*

$$\int_{-1/2+r}^{1/2-r} \frac{\lambda^{2\tau}}{1-2\tau} d\tau \leq \frac{\lambda}{2} \left[\log \frac{1}{r \log \mu} + e^{-1} \right].$$

Proof. Put $l = \log \mu$ and $w = l(\frac{1}{2} - \tau)$. Then $\lambda^{2\tau} = \lambda e^{-w}$, $1 - 2\tau = 2w/l$ and $d\tau = -dw/l$, so the integral equals $\frac{\lambda}{2} \int_{lr}^{l(1-r)} e^{-w} w^{-1} dw \leq \frac{\lambda}{2} \int_{lr}^{\infty} e^{-w} w^{-1} dw \leq \frac{\lambda}{2} [\int_{lr}^1 w^{-1} dw + E_1(1)]$, and $E_1(1) < e^{-1}$. $\square$

6.3. The intermediate range. For Theorem F(b), let $T = \max(T_0, c^3)$. If $c^3 > T_0$, which happens exactly when $\mu > \mu_* = T_0^{1/3}/(2\pi) \approx 2295.4$, the zeros with $T_0 < |t_\rho| \leq T$ are handled by discs of the common radius $r = 1/(11.15x_T)$, $x_T = \log(3T) = \log(3c^3)$. These lie in $|\tau| \leq \frac{1}{2} - r$ by [7], since $1/(5.573412 \log |t_\rho|) > 2r$. Using (3.3) directly for F (which already contains $\hat{b}_d$),

$$\sum_{T_0 < |t_\rho| \leq T} m_\rho |F(\gamma_\rho)|^2 \leq 2n(x_T) (11.15x_T)^2 \lambda j(x_T, \log \mu) m_0(c) B.$$

The factor $\lambda = \sqrt{\mu}$ accounts for possible zeros off the critical line; it is the only source of the extra $\sqrt{\mu}$ in Theorem F(b).

7. PROOFS OF THEOREMS F AND G

Theorem F(a). Take $T = T_0$. The bound is $L_{\text{low}}(c)$ plus the high-zero bound of Section 6, in which the strip integral is bounded by the cruder $L_T(\lambda)$ of Appendix A. By the monotonicity statements of Appendix A, $L_{\text{low}}/(\log \mu)^4$ is maximal at $\mu = 50$ and the high-zero coefficient is maximal at $\mu = 10^4$, which gives $C_{F,a}$ of (A.1). Ball arithmetic gives $C_{F,a} \leq 603317800634135 \leq 6.034 \times 10^{14}$.

Theorem F(b). With $T = \max(T_0, c^3)$,

$$(7.1) \quad P''(\mu) = L_{\text{low}}(c) + M_{\text{mid}}(c) + 2\mathcal{A}(T) \lambda j(x_T, \log \mu) K_*(\mu) + \frac{4\lambda D_+(c)(\log T + 1)}{\pi T},$$

with L_{low} and M_{mid} as in Appendix A ($M_{\text{mid}} = 0$ for $\mu \leq \mu_*$). The four monotonicity statements of Appendix A reduce the envelope to evaluations at $\mu = 50$ and $\mu = \mu_*$:

$$P'' \leq C_L (\log \mu)^4 + C_M \lambda (\log \mu)^5 \mathbf{1}_{\mu > \mu_*} + C_T \lambda (\log \mu)^3 + C_B \lambda \log \mu \leq 5.2 \times 10^{14} \sqrt{\mu} (\log \mu)^5,$$

with $C_L = 395545739509246$, $C_M = 496386197282357$, $C_T = 11498327513$ and $C_B = 3492549$.

Theorem G. By Proposition 3.3(c), $\lambda_{\min} \leq W(\tilde{k})/\|\tilde{k}\|^2 \leq |W(\tilde{k})|/(0.1089H^2)$, and $B \leq U(c)H^2$ with $c^{9/2} = (2\pi)^{9/2} \mu^{9/2}$. The conversion factor is $6621(2\pi)^{9/2}/0.1089 \leq 237522697$, and $237522697 \times 6.034 \times 10^{14} < 1.434 \times 10^{23}$, $237522697 \times 5.2 \times 10^{14} < 1.24 \times 10^{23}$. For (c), with $l = \log \mu$, the ratio to $5.011 \times 10^{17} \mu^8 l^3$ is at most

$$\frac{237522697}{5.011 \times 10^{17}} \left[C_L \frac{l}{\mu^{7/2}} + \frac{C_T}{\mu^3} + \frac{C_B}{\mu^3 l^2} + \mathbf{1}_{\mu > \mu_*} C_M \frac{l^2}{\mu^3} \right].$$

The first three terms decrease for $\mu \geq 50$ and the last for $\mu \geq \mu_*$, so the sum of their values at 50 (first three) and at μ_* (last) bounds the ratio; ball arithmetic gives $0.831028501 \dots < 0.832$.

8. NUMERICAL EVIDENCE

	$c = 10\pi$	$c = 14\pi$	$c = 20\pi$
out-of-band mass ratio, $n = 0$, one factor of $1 - E_L$	1.191	1.142	1.102
out-of-band mass ratio, $n = 0$, L	1.481	1.359	1.260
out-of-band mass ratio, $n = 4$, L	1.299	1.254	1.204

TABLE 2. Normalized out-of-band mass $B(f)/\|f\|^2$ of $f = \psi_n(1 - E_L)$ or $f = \psi_n L$ on $[-1, 1]$ (bandwidth c), divided by $1 - \lambda_n = B(\psi_n)/\|\psi_n\|^2$. Computed with 60 digits as $\|f\|^2$ minus the in-band mass.

μ	$W(k)/B$	$W(\tilde{k})/\tilde{B}$	$\ k\ ^2/H^2$	$\ \tilde{k}\ ^2/\ \tilde{h}\ ^2$
5	1.623	2.141	0.222685	0.222685
7	1.805	2.292	0.221439	0.221439

TABLE 3. Galerkin values ($N = 140$ Legendre modes, 480 bits); not part of any proof. Here $\tilde{B} = \tilde{I}_4^2 b_0 + \tilde{I}_0^2 b_4$ is the out-of-band mass of the unsmoothed combination with the smoothed coefficients; it differs from B by a relative amount of order $\eta(c)\sqrt{U(c)}$.

- For the unsmoothed vector k and its trace $J = e(-a)$, $|\hat{k}(t_\rho)|^2/(J^2/(\frac{1}{4} + t_\rho^2))$ is $1.5 \times 10^{-5} - 2.0 \times 10^{-3}$ at the first eight zeros for $\mu = 5, 7, 10$, and the partial zero sum over the first 40 zeros is concentrated at $t_\rho \approx 2.5c - 3c$.
- For the smoothed vector, $J^2/B(\tilde{h}) = 7.17, 13.9, 20.2$ at $\mu = 5, 7, 10$.
- The bound of Lemma 4.1 was compared with direct quadrature of $\hat{g}_{0,L}(\lambda t)$ at $\mu = 5$ for $t/c \in \{1, 2, 5, 10, 20, 40, 80\}$; the ratio actual/bound stays in $[3 \times 10^{-7}, 9 \times 10^{-7}]$.

APPENDIX A. EXPLICIT FUNCTIONS AND MONOTONICITY

Throughout, $l = \log \mu$, $\mu \geq 50$, $c = 2\pi\mu$, $\lambda = \sqrt{\mu}$, $a = \log \lambda$, $T_0 = 3 \cdot 10^{12}$, $R = c^3$, $L_R = \log(2R)$ and $d = c^2$.

Prolate side.

$$z = \frac{32e^8 \sqrt{8/\pi}}{\ell}, \quad V(c) = (2 + 48e^8) \sqrt{16\pi/\ell} + \frac{2z}{\sqrt{\ell}} \left(\frac{4}{c} + \frac{2}{\sqrt{3}} \right),$$

$$E_J(c) = V(c) + 1.78\sqrt{8} \left[\frac{\pi}{2} + 1 + 2 \log \frac{c}{3} + 5.49 \right] + \frac{\sqrt{2.0408}}{2},$$

$D_J(c) = [\sqrt{8c} + E_J(c)]^2 \leq D_+(c)$, $m_0(c)$ as in (3.3), $Z_{\text{raw}} = [\sqrt{m_0} + \sqrt{D_J/c}]^2$, $Z = Z_C l^2$, $K_*(\mu)$ as in Proposition 5.3, and

$$Y = [\sqrt{Z} + \sqrt{2}(a+4)(\sqrt{K_*} + \frac{1}{4}\sqrt{Z})]^2, \quad \mathfrak{A}_0 = L_R Z + K_*/R^2, \quad \mathfrak{A}_1 = 2(a+4)^2 L_R Z + Y/R^2.$$

Zero side. With $C_N = 3.325 + 0.2/e$,

$$S(d) = \frac{\frac{4}{3} \log d + 1}{\pi d} + \frac{0.772 \log d + 2C_N + 0.193}{d^2},$$

$$n(x) = \left(\frac{1}{4\pi} + 0.222 \right) x + 0.55 \log x + 4.9 + \frac{0.4}{e}, \quad j(x, l) = \frac{1}{2} \left[\log \frac{11.15x}{l} + e^{-1} \right],$$

$\mathcal{A}(T)$ as in Section 6, $x_0 = \log(3T_0)$, $L_T(\lambda) = \frac{1}{2} + \frac{\lambda}{2} \log \frac{11.15x_0}{2}$ (the strip bound of [5, proof of Lemma 5.1]), $H_T = (\log T_0 + 1)/(\pi T_0)$, and

$$L_{\text{low}}(c) = 48\pi[\mathfrak{A}_0 + 2\sqrt{\mathfrak{A}_0\mathfrak{A}_1}] + 4D_+(c)S(c^2),$$

$$M_{\text{mid}}(c) = \begin{cases} 0, & c^3 \leq T_0, \\ 2n(x)(11.15x)^2\lambda j(x, l)m_0(c), & x = \log(3c^3), \quad c^3 > T_0. \end{cases}$$

Monotonicity. The functions named in the four items below are sums or products of positive monotone functions. All are decreasing on the stated ranges, except the high-zero coefficient in item 2, which is increasing. The derivatives of $n(x)/x$ and $j(x, l)/x$ are checked in `all_mu_identities.py`.

- (1) Z_{raw}/l^2 , $\sqrt{K_*}/(c^3\sqrt{Z})$, $\mathfrak{A}_0/l^3$, $\mathfrak{A}_1/l^5$ and $D_+(c)S(c^2)$ decrease for $\mu \geq 50$. At $\mu = 50$, $K_*/(R^2Z) < 0.000033$, so $\mathfrak{A}_0 \leq 5Z_C l^3$, $\mathfrak{A}_1 \leq 12(1.53)^2 Z_C l^5$ and $48\pi[\mathfrak{A}_0 + 2\sqrt{\mathfrak{A}_0\mathfrak{A}_1}] \leq 48\pi \cdot 26Z_C l^4$.
- (2) For $T = T_0$ the high-zero coefficient $2\mathcal{A}(T_0)L_T(\lambda)K_*(\mu) + 4\lambda D_+(c)H_T$ increases in μ . Hence, for $50 \leq \mu \leq 10^4$,

$$(A.1) \quad \frac{|W(\tilde{k})|}{B l^4} \leq C_{F,a} := 48\pi \cdot 26Z_C + \frac{4D_+(100\pi)S((100\pi)^2) + 2\mathcal{A}(T_0)L_T(100)K_*(10^4) + 400D_+(2\pi \cdot 10^4)H_T}{(\log 50)^4}.$$

- (3) For $\mu > \mu_*$ (where $x = x_T = \log(3c^3)$), x/l , $n(x)/l$, $j(x, l)$ and m_0/l^2 decrease. Hence $M_{\text{mid}}/(\lambda l^5)$ is bounded on $\mu > \mu_*$ by its right limit at μ_* , computed from the formula for $c^3 > T_0$ (note $M_{\text{mid}}(\mu_*) = 0$); this gives C_M .
- (4) K_*/c^4 decreases, $T^2 \geq T_0^{2/3}c^4$ and $x_T \leq (x_0/\log 50)l$ (since x_T/l decreases), which give C_T ; $\log T + 1 \leq x_T$, $T \geq c^3$ and the decrease of $D_+(c)/c^3$ give C_B ; and $L_{\text{low}}/l^4 \leq C_L$ by item 1.

REPRODUCIBILITY

All computer-assisted steps are reproducible from the public repository <https://github.com/moriryota/prolate-weil-bounds>, archived on Zenodo, doi:10.5281/zenodo.23092422 (all versions; this paper corresponds to release v1.1.0). It contains every script and its raw output. The scripts in `proofs-III/` use ball arithmetic (Arb, via python-flint) or SymPy; every printed upper bound is rounded upward, every lower bound downward, and each is asserted against the ball.

Statement	Script
Lemma 3.1, (3.2), Proposition 3.3, (3.3)	<code>proofs-III/layer_constants.py</code>
Lemma 3.2 (2.84278; key <code>endpoint_coeff</code>)	<code>proofs-III/theoremF_a_constants.py</code>
Lemma 5.1, zero-count constants of Section 6	<code>proofs-III/trace_constants.py</code>
Lemma 4.1 (identities)	<code>proofs-III/cancellation_identities.py</code>
Lemma 4.1, Propositions 4.2, 5.3; Theorems F(a), G(a)	<code>proofs-III/theoremF_a_constants.py</code>
P'' , Theorems F(b), G(b), (c)	<code>proofs-III/theoremF_b_constants.py</code>
Sections 6–7 (identities)	<code>proofs-III/all_mu_identities.py</code>

The numerical values in Section 8 and the check of Lemma 6.1 come from `numerics-III/`; they are not part of any proof.

USE OF GENERATIVE AI

This work was carried out with extensive use of generative AI systems, namely Anthropic Claude, Google Gemini and OpenAI GPT. Under the direction of the author, these systems drafted the proofs, implemented and ran the computations, reviewed one another's drafts adversarially, and

drafted this manuscript. The author chose the problem and the strategy, assigned and reviewed the tasks, and takes full responsibility for the content. The AI systems are not authors of this paper.

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